

Abin Anil

AI/ML Engineer | Python, GenAI, RAG & Agentic Systems

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Professional Summary

AI/ML Engineer with a strong Python foundation, building and deploying GenAI, LLM, and agentic AI applications using LangChain, LangGraph, Hugging Face Transformers, RAG pipelines, and vector search (PostgreSQL/pgvector). Hands-on across the full AI development lifecycle – machine learning theory, deep learning, NLP, model fine-tuning (QLoRA/PEFT), multi-agent orchestration, backend APIs, and cloud deployment – backed by 2+ years of enterprise production-engineering experience (REST APIs, SQL, observability, root-cause analysis) that translates directly into building AI systems that are reliable and production-ready, not just proof-of-concept notebooks.

Technical Skills

Programming: Python, SQL, JavaScript

Machine Learning: Supervised, Unsupervised & Semi-Supervised Learning, Regression, Classification, Ensemble Methods (XGBoost, Random Forest), Feature Engineering, Hyperparameter Tuning, Model Evaluation (Precision/Recall/F1), Scikit-Learn

Deep Learning: Neural Networks, Backpropagation, CNNs, RNNs/LSTMs, Transformer Architectures, Reinforcement Learning (fundamentals), TensorFlow, PyTorch

GenAI/LLM & Agentic AI: Agentic AI Workflows, Multi-Agent Systems & Orchestration, Workflow Orchestration (LangGraph state graphs), Agent Memory & State Management, Tool Calling / Function Calling (Model Context Protocol – MCP), Retrieval-Augmented Generation (RAG), LLM Application Development & System Integration, Generative AI (GANs, VAEs, Diffusion Models, Transformers), Generative AI Agents, Large Language Modeling, LLM Architecture, OpenAI API Integration, LangChain, LangGraph, Hugging Face, Prompt Engineering, Prompt Patterns, Context Engineering, Vector Embeddings, Semantic Search, QLoRA, PEFT, Unsloth, 4-bit Quantization

NLP: NLTK, spaCy, Text Preprocessing, Tokenization, TF-IDF, Text Representation, Language Models, Sentiment Analysis, Text Classification

Backend/Data: FastAPI, Flask, Streamlit, PostgreSQL, SQL & NoSQL, pgvector, SQLAlchemy, psycopg, REST APIs, Pandas, NumPy, Matplotlib, Statistics & Probability

Cloud/Deployment: Docker, Docker Compose, GCP Cloud Run, Jenkins CI/CD, Git, GitHub

Engineering Fundamentals: API Debugging, Root Cause Analysis, Observability, Incident Management, OOP, Data Structures & Algorithms, SDLC

Projects

Autonomous Multi-Agent Trading Architecture (Nexus Intelligence)

Sep 2026

GitHub: <https://github.com/abinanil000datascience/nexus-ai-trading-engine>

- Architected a decoupled, asynchronous multi-agent trading system, replacing a single-threaded Streamlit prototype, by rebuilding the backend on FastAPI and orchestrating four agents (Scout, Quant, Analyst, Gatekeeper) with LangGraph state graphs and persistent agent memory.
- Implemented structured tool calling between agents via the Model Context Protocol (MCP), letting agents invoke shared tools and pass state through a coordinated workflow.
- Enabled sentiment-aware, real-time trading verdicts by combining a deterministic Quant agent – computing live 14-day RSI/MACD on streaming yfinance data via Pandas – with an NLP-driven Analyst agent that uses Gemini 3.5 Flash to classify live financial news as Bullish/Bearish/Neutral.
- Eliminated critical concurrency failures in the agent orchestration layer, resolving asyncio event-loop collisions between FastAPI and LangGraph execution threads, by restructuring API endpoints to run agent workflows on isolated background threads.
- Delivered a real-time monitoring dashboard (React, Vite, Tailwind CSS v4, Recharts) showing 60-day price action, live agent telemetry, and trade verdicts from the FastAPI backend.

Tech Stack: Python, FastAPI, LangGraph, MCP, Gemini 3.5 Flash, yfinance, Pandas, asyncio, React, Vite, Tailwind CSS, Recharts.

Autonomous Healthcare ITSM Resolution Engine

Aug 2026

GitHub: https://github.com/abinanil000datascience/Healthcare_ITSM_Agent

- Built an LLM-powered IT ticket resolution system by fine-tuning a local Meta Llama-3 (8B) model with Unsloth and QLoRA (4-bit quantization) for domain-specific ticket classification and generation, running as a zero-marginal-cost local inference pipeline with no paid LLM API.
- Implemented a full Retrieval-Augmented Generation (RAG) pipeline, combining fine-tuned Llama-3 generation with hybrid PostgreSQL/pgvector retrieval to ground ticket responses in historical resolution data.
- Accelerated retrieval across a historical ticket corpus by designing a sub-linear hybrid search architecture in PostgreSQL/pgvector, combining keyword matching (tsvector/BM25) with HNSW-indexed semantic vector search.
- Deployed the retrieval system as an asynchronous FastAPI backend using sentence-transformers, fully containerized with Docker.

Tech Stack: Meta Llama-3 (8B), Unsloth, QLoRA, Hugging Face, PostgreSQL, pgvector, HNSW, sentence-transformers, FastAPI, Docker.

Enterprise AI Career Platform

Aug 2026

GitHub: <https://github.com/abinanil000datascience/ai-career-platform>

- Built an AI career platform for resume analysis and applicant tracking, automating resume-to-role matching with a semantic ATS evaluation engine using LangGraph workflow orchestration and vector embeddings, backed by a PostgreSQL and pgvector data layer for retrieval-grounded matching.
- Integrated multiple LLM providers (Gemini API, Groq API) on an asynchronous FastAPI backend, containerized and deployed on GCP Cloud Run.

Tech Stack: Python, FastAPI, LangGraph, Gemini API, Groq API, PostgreSQL, pgvector, Docker, GCP Cloud Run.

AI-Powered Supply Chain Risk Predictor

May 2026

GitHub: <https://github.com/abinanil000datascience/supply-chain-risk-predictor>

- Forecasted freight delays on a 97% on-time, class-imbalanced dataset with a weighted Scikit-Learn/XGBoost pipeline, serving real-time delay-risk predictions through a Django REST API with a React front end.

Tech Stack: Python, Scikit-Learn, XGBoost, Django, React.

Professional Experience

Technical Support Engineer, Feathersoft

Aug 2024 – Present

Distributed DNA diagnostics platform (DDC)

- **API & Database Operations:** Managed production reliability for a distributed DNA diagnostics platform (DDC), ensuring data integrity across complex probability (PI) calculations and automated report generation workflows, sustaining 99.8% system uptime.
- Reduced production error rates by 22% by debugging API-layer failures and data-ingestion bottlenecks through network traffic inspection, cURL requests, and payload validation using Postman and Swagger.
- Spearheaded system observability and anomaly detection by analyzing application logs in Kibana and Grafana to isolate root causes of complex workflow disruptions, accelerating incident resolution time by 15%.
- Reduced Mean Time to Resolution (MTTR) by 20% by executing direct SQL (SELECT/UPDATE) database interventions to resolve live data discrepancies and maintain the structural integrity of diagnostic reporting.
- Engineered and validated temporary diagnostic workflows within staging and test environments, acting as a technical bridge between end-users and the core development team to unblock production.
- Led cross-functional incident triage with US-based enterprise clients and engineering teams, translating complex diagnostic logs into actionable technical fixes, sustaining a 99% CSAT rating across 120+ escalated incidents. Recognized as **Employee of the Month (January 2026)** for technical excellence and client communication.

Professional Training

Data Science With AI Professional Certification – MetaLearn

Nov 2025 – Jun 2026

Structured part-time program, completed alongside full-time role

- **ML, Deep Learning & NLP:** Studied supervised/unsupervised/semi-supervised learning, backpropagation, CNNs, RNNs/LSTMs (TensorFlow, PyTorch), text representation, and language models, on a Math & Statistics foundation.
- **GenAI Engineering:** Covered LLM architecture, GANs/VAEs/diffusion models, prompt engineering, RAG and LLM pipelines, and OpenAI API integration into backend applications.
- **Tooling:** Python for Data Science, SQL & NoSQL, Flask, Streamlit, GitHub Copilot.

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Dec 2020 – May 2024

University College of Engineering, Thodupuzha

CGPA: 7.33/10.0

Certifications & Training

- **Fundamentals of AI Agents Using RAG and LangChain – IBM** Sep 2026
ID: S0HTSWVNLSTW | [Verify Credential](#)
- **Python Programming & Machine Learning Intensive Program – Reverttech IT Solutions** Dec 2024
- **Advanced Machine Learning Workshop – Techmaghi Training Institute** Sep 2022